

Identifiable Disentangled Representation Learning for Causal Inference under Network Interference

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Abstract

Estimating individual treatment effects (ITE) from networked observational data is a fundamental but challenging task. Network connections not only introduce complex confounding bias but also generate spillover effects, making causal inference in such settings particularly difficult. To tackle these challenges, existing methods often attempt to learn confounders from all observed variables to predict potential outcomes. However, this approach overlooks heterogeneous latent factors that differentially influence treatment assignment and outcomes, which may prevent accurate identification of the confounders and reduce the effectiveness of bias correction. To overcome these limitations, we propose the Network Disentangled Identifiable Variational Autoencoder (NDiVAE), a structured framework that learns three distinct latent factors for each unit, including instrumental, confounding, and adjustment factors. NDiVAE further incorporates graph-based neighborhood information to learn aggregated factor representations, applies sample reweighting to mitigate confounding bias, and enforces causal regularization to ensure precise disentanglement of latent factors. We establish theoretical guarantees through the identifiability of these latent factors and derive a generalization error bound for ITE estimation. Extensive experiments on synthetic and semi-synthetic datasets demonstrate that NDiVAE consistently outperforms state-of-the-art methods in estimating treatment effects.

CCS Concepts

• **Computing methodologies** → **Causal reasoning and diagnostics**; • **Mathematics of computing** → **Causal networks**.

Keywords

Causal inference; individual treatment effect; networked observational data; disentanglement

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1 Introduction

Estimating individual treatment effects (ITE) is a central task in causal inference, supporting causal decision-making in healthcare [37], education [8], economics [40], and public policy [23]. In many real-world applications, observational units are connected through networks, which has led to increasing interest in causal inference on networked observational data. The presence of network structures introduces new and challenging confounding bias due to structural dependencies among units [16]. More importantly, in networked environments, the treatment received by one unit can influence the outcomes of its neighboring units. This phenomenon, known as network interference or spillover effects [10], further violates the Stable Unit Treatment Value Assumption (SUTVA) [32] on which many causal inference methods rely. These challenges motivate the development of causal inference methods tailored to networked data.

Several studies have extended traditional ITE estimation methods designed for independent data to accommodate network interference [25, 30, 31]. For example, Forastiere et al. [10] generalized the unconfoundedness assumption to settings with network interference, defined three types of effects, including main effect, spillover effect, and total effect, and established the identification of network effects using a joint propensity score approach. NetEst [16] employed graph neural networks with representation balancing to handle relational dependencies in treatment assignment and outcomes. Cai et al. [4] proposed theoretical generalization bounds and constructed conditional outcome estimators through semiparametric regression to improve causal effect estimation under interference.

Despite these advances, existing methods fail to account for different types of latent factors that influence treatment assignment and outcome generation in distinct ways. In practice, latent factors can play fundamentally different roles. For example, in personalized medicine, physicians seek to identify the treatment that offers the greatest benefit (outcome) to an individual patient. The patient's socioeconomic status may affect both treatment and outcome (confounding factors), physician prescribing preferences may affect only treatment (instrumental factors), and genetic predispositions may affect only outcomes (adjustment factors). Figure 1 illustrates the causal structure of a two-node network, highlighting dependencies among key variables under network interference. Collapsing these heterogeneous influences into a single representation can create spurious dependencies and reduce the accuracy of causal effect estimation. As a result, traditional methods struggle to accurately identify confounders, which in turn undermines reliable potential

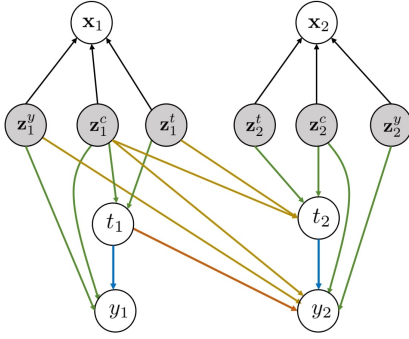


Figure 1: Causal graph of a two-node network. x denotes observed features (proxy variables), t the treatment, and y the outcome. Latent factors z_t , z_c , and z_y represent instrumental, confounding, and adjustment factors. Black edges indicate the generation of proxy variables; green edges show effects of individual latent factors on their own treatment and outcome; yellow edges depict their influence on neighbors through the network. Blue edges denote a unit's treatment effect on its outcome, while red edges represent spillover effect from neighbors. Node 2 is shown as an example, with spillover effects restricted to first-order neighbors.

outcome estimation and causal inference. Furthermore, most approaches address confounding bias by minimizing differences in confounder representations between treated and control groups [11, 14, 17]. However, in real-world scenarios, the feature distributions of these groups often differ significantly. Enforcing strict balance between group-specific confounder representations can degrade the effectiveness of outcome prediction. This motivates the need for more flexible strategies to mitigate confounding bias without sacrificing predictive accuracy.

To address these challenges, we propose the Network Disentangled Identifiable Variational Autoencoder (NDiVAE), a structured framework designed to learn three distinct latent factors and to effectively reduce confounding bias. Our framework consists of three main components. First, the causal representation module learns three separate latent factors for each node, corresponding to instrumental, confounding, and adjustment factors. This is achieved by using an identifiable variational autoencoder combined with causal regularization, guiding the latent space toward the desired factorization. Second, in the outcome prediction module, information from neighboring nodes is incorporated to enhance potential outcome estimation. Three dedicated graph neural networks are employed to learn the network-integrated representations corresponding to the three latent factors. We derive a generalization error bound for ITE estimation and use an Integral Probability Metric (IPM) to capture confounding bias, which is then mitigated by reweighting samples. Inspired by this bound, we adopt a weighted regression objective for potential outcome estimation. Finally, the causal regularization module imposes auxiliary constraints to align the learned representations with the assumed causal structure. With these components, the model can predict potential outcomes and enable accurate ITE estimation. Our main contributions can be summarized as follows:

- We investigate individual treatment effect estimation in networked observational data with interference, emphasizing the necessity of explicitly disentangling instrumental, confounding, and adjustment factors.
- We propose a novel framework, NDiVAE, which learns three latent factors for each unit via an identifiable variational autoencoder, and integrates graph aggregation and sample reweighting to mitigate confounding bias under interference while preserving outcome prediction accuracy. Theoretical guarantees are provided through latent factor identifiability and a generalization error bound for treatment effect estimation.
- We conduct extensive experiments on both synthetic and semi-synthetic real-world datasets, demonstrating significant improvements over state-of-the-art methods.

2 Preliminary

In this section, we provide a brief overview of the technical preliminaries. We first introduce the notations used throughout the paper and then present formal definitions of causal effects for networked observational data under network interference.

Networked Observational Data. We consider a graph $G = \langle A, X \rangle$, where X is the feature matrix containing covariates for n nodes, and each row x_i represents the feature vector of node i . The adjacency matrix $A \in \{0, 1\}^{n \times n}$ characterizes the pairwise relationships among nodes, where $A_{ij} = A_{ji} = 1$ indicates an undirected edge between nodes i and j ; otherwise, $A_{ij} = 0$. We restrict our focus to binary treatments, i.e., $t_i \in \{0, 1\}$. The treatment assignments are then represented by a binary vector $T = [t_1, \dots, t_n]$, where $t_i = 1$ if node i receives the treatment and $t_i = 0$ otherwise. The observed outcomes are denoted by $Y = [y_1, \dots, y_n]$, where each $y_i \in \mathbb{R}$ represents the outcome of node i . We assume that the outcomes are continuous.

Learning Causal Effects. Throughout this work, we adopt the potential outcomes framework to evaluate treatment effects [33]. For the entire population, the potential outcome is defined as $Y(T)$, representing the vector of outcomes that would be observed if the treatment assignment were set to T .

Next, following [1, 11], we define the global causal effect over the entire graph, denoted by $\tau(X)$, as the difference in the expected potential outcomes of all units under two alternative treatment assignment vectors, T' and T''

$$\tau(X) = \mathbb{E}[Y(T')|X, A] - \mathbb{E}[Y(T'')|X, A]. \quad (1)$$

Estimating this global quantity requires computing the individual treatment effect for each node i , denoted by $\tau(x_i)$. In networked settings, the potential outcome of a node depends not only on its own features and treatment assignment, but also on the features $\{x_j\}_{j \in \mathcal{N}_i}$ and treatment assignments $\{t_j\}_{j \in \mathcal{N}_i}$ of its neighbors, where $\mathcal{N}_i$ denotes the set of one-hop neighbors of node i . Furthermore, to capture interference from neighbors' treatments $\{t_j\}_{j \in \mathcal{N}_i}$, we introduce a summary function $G : 2^T \rightarrow [0, 1]$, following the formulation in [10], that aggregates their influence. Specifically, we define the *peer exposure* $g_i = G(\{t_j\}_{j \in \mathcal{N}_i})$ as the ratio of treated neighbors, that is, $g_i = \frac{\sum_{j \in \mathcal{N}_i} t_j}{|\mathcal{N}_i|}$, which naturally takes values in the interval $[0, 1]$. We explicitly assume that such peer exposure captures the key interference effect relevant to each unit's outcome.

Formally, to emphasize the factors influencing potential outcomes under network interference, we reformulate the potential outcome as $y_i(t_i, g_i)$, which represents the outcome of unit i given its own treatment $t_i \in \{0, 1\}$ and a continuous peer exposure level g_i [16]. The individual treatment effect for unit i is then defined as:

$$\tau_i = \mathbb{E}[y_i(t'_i, g'_i) | \mathbf{x}_i, \{\mathbf{x}_j\}_{j \in N_i}] - \mathbb{E}[y_i(t''_i, g''_i) | \mathbf{x}_i, \{\mathbf{x}_j\}_{j \in N_i}]. \quad (2)$$

This expression quantifies the expected outcome difference for unit i under two distinct treatment and peer exposure configurations, highlighting a key distinction between networked and independent settings, namely, that neighboring features $\{\mathbf{x}_j\}_{j \in N_i}$ and peer exposure g_i jointly influence outcomes and induce network interference.

Building on Eq. (2), we examine several network-level causal effects. Following [5, 10], we focus on three commonly studied effect types under network interference:

Definition 2.1 (Individual Main Effect (IME)). The IME measures the effect of unit i 's own treatment while fixing peer exposure at g_i : $\tau^{(1, g_i), (0, g_i)}(x_i) := \mathbb{E}[y_i(1, g_i) | \mathbf{x}_i, \{\mathbf{x}_j\}_{j \in N_i}] - \mathbb{E}[y_i(0, g_i) | \mathbf{x}_i, \{\mathbf{x}_j\}_{j \in N_i}]$.

Definition 2.2 (Individual Spillover Effect (ISE)). The ISE quantifies network interference by capturing how neighbors' treatments affect unit i while its own treatment is held fixed:

$$\tau^{(t_i, g_i), (t_i, 0)}(x_i) := \mathbb{E}[y_i(t_i, g_i) | \mathbf{x}_i, \{\mathbf{x}_j\}_{j \in N_i}] - \mathbb{E}[y_i(t_i, 0) | \mathbf{x}_i, \{\mathbf{x}_j\}_{j \in N_i}].$$

Definition 2.3 (Individual Total Causal Effect (ITCE))¹. The ITCE reflects the combined impact of own treatment and peer exposure: $\tau^{(1, g_i), (0, 0)}(x_i) := \mathbb{E}[y_i(1, g_i) | \mathbf{x}_i, \{\mathbf{x}_j\}_{j \in N_i}] - \mathbb{E}[y_i(0, 0) | \mathbf{x}_i, \{\mathbf{x}_j\}_{j \in N_i}]$.

Once unit-level ITCEs are estimated, the average treatment effect (ATCE) is computed as $\text{ATCE} = \frac{1}{n} \sum_{i=1}^n \tau^{(1, g_i), (0, 0)}(x_i)$. The average main effect (AME) and average spillover effect (ASE) are defined analogously and omitted for brevity.

To estimate the individual treatment effect defined in Eq. (2), both potential outcomes under the two corresponding treatment conditions must be considered. However, for each unit, only one of these outcomes is observable; the other remains unobserved and is referred to as the *counterfactual outcome*. Consequently, a central challenge in ITE estimation lies in accurately recovering this missing counterfactual outcome.

Assumptions. To ensure that potential outcomes can be inferred from networked observational data, we introduce the following assumptions [10].

Assumption 1 (Consistency). The observed outcome corresponds to the potential outcome under the realized treatment and peer exposure, i.e., if unit i receives treatment t_i and experiences peer exposure g_i , then the observed outcome satisfies $y_i = y_i(t_i, g_i)$.

Assumption 2 (Positivity). Each unit has a positive probability of receiving any treatment and peer exposure level given its own and neighbors' features, i.e., $0 < P(t_i, g_i | \mathbf{x}_i, \{\mathbf{x}_j\}_{j \in N_i}) < 1$.

Assumption 3 (Neighborhood interference). Unit i 's potential outcome depends only on its own treatment t_i and a summary exposure $g_i(\{t_j\}_{j \in N_i})$ from its neighbors, and is unaffected by treatments of non-neighbors or by different neighbor assignments yielding the same g_i . Formally, if $g_i(\{t_j\}_{j \in N_i}) = g_i(\{t'_j\}_{j \in N_i})$, then $y_i(t_i, \{t_j\}_{j \in N_i}, \{t'_j\}_{j \in N_i \cup \{i\}}) = y_i(t_i, \{t'_j\}_{j \in N_i}, \{t'_j\}_{j \in N_i \cup \{i\}})$.

¹To distinguish it from the standard individual treatment effect (ITE), we denote this quantity as ITCE.

3 Methodology

In this section, we present a novel framework, *NDiVAE*, for estimating individual treatment effects from networked observational data. Guided by the causal structure illustrated in Figure 1, the corresponding model architecture is shown in Figure 2.

3.1 Causal Representation Module

Traditional causal inference methods often treat all observed covariates as confounders, using reweighting, matching, or balanced representation learning to mitigate bias [11, 15, 17]. However, uniformly treating covariates as confounders can obscure their distinct causal roles and limit accurate treatment effect estimation [2, 22, 38]. Disentangled representation learning addresses this by separating latent factors, improving outcome prediction. However, existing disentangled approaches have not been extended to settings with networked observational data under interference. Motivated by this gap, we follow insights from prior work and assume that the observed features $\mathbf{x}$ serve as proxy variables for three distinct types of latent factors: instrumental factors $\mathbf{z}_t$, confounding factors $\mathbf{z}_c$ and adjustment factors $\mathbf{z}_y$.

To enable disentangled representation learning from proxy variables, we adopt the identifiable variational autoencoder (iVAE) framework [19], which ensures the identifiability of latent representations under mild conditions. Following the iVAE formulation, we introduce the auxiliary observed variable $\mathbf{u} \in \mathbb{R}^m$. The generative model is factorized as

$$\begin{aligned} p(\mathbf{x}, \mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y | \mathbf{u}) &= p(\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y | \mathbf{u}) p(\mathbf{x} | \mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y) \\ &= p(\mathbf{z}_t | \mathbf{u}) p(\mathbf{z}_c | \mathbf{u}) p(\mathbf{z}_y | \mathbf{u}) p(\mathbf{x} | \mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y). \end{aligned} \quad (3)$$

Here, the conditional prior $p(\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y | \mathbf{u})$ assumes conditional independence across different types of latent factors and is modeled using Gaussian distributions from the exponential family. The priors for the instrumental, confounding, and adjustment factors are given by:

$$p(\mathbf{z}_i | \mathbf{u}) = \prod_{d=1}^{D_i} \mathcal{N}(z_{i,d} | \hat{\mu}_{i,d}(\mathbf{u}), \hat{\sigma}_{i,d}^2(\mathbf{u})), \quad i \in \{t, c, y\}, \quad (5)$$

where D_t, D_c, D_y denote the dimensionalities of $\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y$, respectively, and the mean and variance functions $\hat{\mu}_{i,d}(\cdot)$ and $\hat{\sigma}_{i,d}^2(\cdot)$ are parameterized by separate multilayer perceptrons (MLPs).

For continuous proxy variables $\mathbf{x}$, the likelihood is formulated as

$$p(\mathbf{x} | \mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y) = \prod_{j=1}^{D_x} \mathcal{N}(x_j | \hat{\mu}_{x,j}(\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y), \hat{\sigma}_{x,j}^2(\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y)), \quad (6)$$

where D_x is the dimensionality of $\mathbf{x}$, and $\hat{\mu}_{x,j}(\cdot)$, $\hat{\sigma}_{x,j}^2(\cdot)$ are implemented by MLPs.

For inference, the variational posterior is also factorized and modeled separately for each latent variable:

$$q(\mathbf{z}_i | \mathbf{x}, \mathbf{u}) = \prod_{d=1}^{D_i} \mathcal{N}(z_{i,d} | \hat{\mu}'_{i,d}(\mathbf{x}, \mathbf{u}), \hat{\sigma}'_{i,d}{}^2(\mathbf{x}, \mathbf{u})), \quad i \in \{t, c, y\}, \quad (7)$$

where $\hat{\mu}'_{i,d}(\cdot)$ and $\hat{\sigma}'_{i,d}{}^2(\cdot)$ are parameterized by MLPs that jointly take $\mathbf{x}$ and $\mathbf{u}$ as inputs.

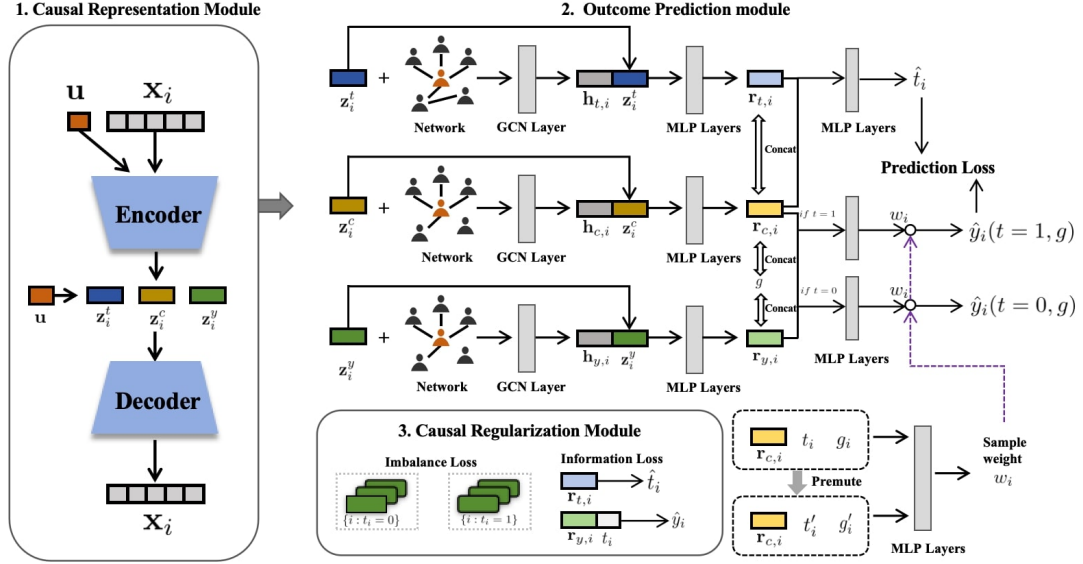


Figure 2: An illustration of the proposed framework NDIVAE.

3.2 Outcome Prediction Module

Given the latent factor representations learned in the previous section, we now aim to leverage them to predict potential outcomes.

3.2.1 Aggregated Representation. Relying only on individually learned latent factors can be inaccurate because a unit's outcome is influenced by its neighbors. To address this, we aggregate each unit's neighbors' representations and combine them with its own latent factors to form network-integrated representations.

Following extensive prior work on network representation learning, we adopt Graph Convolutional Networks (GCN) [5, 16, 21] to integrate information from the local neighborhood. Specifically, let $\mathbf{z}_{c,i} \in \mathbb{R}^{D_c}$ denote the latent confounding factors of unit i . The aggregation is given by

$$\mathbf{h}_{c,i} = \sigma \left(\sum_{j \in \mathcal{N}_i \cup \{i\}} \frac{1}{\sqrt{\deg(i) \deg(j)}} \mathbf{W}^\top \mathbf{z}_{c,j} \right) \in \mathbb{R}^{D_{hc}}, \quad (8)$$

where $\mathcal{N}_i$ denotes the set of first-order neighbors of node i , $\mathbf{W} \in \mathbb{R}^{D_c \times D_{hc}}$ is a learnable weight matrix, $\deg(i)$ is the degree of node i , and $\sigma(\cdot)$ is a non-linear activation. The node's own latent factor and its aggregated neighborhood representation are then concatenated and passed through a transformation function to produce the *network-integrated representation*:

$$\mathbf{r}_{c,i} = f_r(\mathbf{z}_{c,i}, \mathbf{h}_{c,i}) \in \mathbb{R}^{D_{rc}}, \quad (9)$$

where $f_r(\cdot)$ denotes a MLP.

The latent factors $\mathbf{z}_{t,i} \in \mathbb{R}^{D_t}$ and $\mathbf{z}_{y,i} \in \mathbb{R}^{D_y}$, are aggregated in the same manner as Eq. (8) with corresponding network-integrated representations $\mathbf{r}_{t,i}$ and $\mathbf{r}_{y,i}$ defined analogously to (9). By incorporating aggregated neighbor information for $\mathbf{z}_{t,i}$, $\mathbf{z}_{c,i}$, and $\mathbf{z}_{y,i}$, our model captures both intrinsic and neighborhood factors for each unit.

3.2.2 Potential Outcome Estimation. After aggregating the latent confounding factors from both the individual unit and its neighbors, our model obtains the network-integrated representation $\mathbf{r}_{c,i}$. This aggregated confounder representation is critical because, in networked observational settings, treatment assignment t_i , peer exposure g_i , and confounders $\mathbf{z}_{c,i}$ (including those of neighbors) are typically correlated, as illustrated in Figure 1. Such correlations can lead to biased estimates of potential outcomes if not properly addressed.

To mitigate confounding bias, we construct a *calibration distribution* that serves as an ideal reference in which (t_i, g_i) and $\mathbf{r}_{c,i}$ are independent while preserving their marginals [41], i.e., $p_{\text{cal}}(t_i, g_i, \mathbf{r}_{c,i}) = p(t_i, g_i)p(\mathbf{r}_{c,i})$. In practice, we approximate p_{cal} by constructing a calibrated dataset through independently permuting (t_i, g_i) and $\mathbf{r}_{c,i}$ across the original samples, thereby breaking joint dependencies while retaining their marginal distributions. We then reweight the observed samples to approximate this calibrated distribution by estimating the density ratio between the observed and calibrated datasets:

$$w_i = \frac{p_{\text{cal}}(t_i, g_i, \mathbf{r}_{c,i})}{p_{\text{obs}}(t_i, g_i, \mathbf{r}_{c,i})} = \frac{p(t_i, g_i)p(\mathbf{r}_{c,i})}{p(t_i, g_i, \mathbf{r}_{c,i})}. \quad (10)$$

Since direct density estimation in high dimensions is challenging, we employ a binary classifier to distinguish samples from the observed distribution p_{obs} and the calibration distribution p_{cal} . Using the density ratio trick, the classifier's output probabilities provide an efficient estimate of w_i . These weights are then incorporated into the outcome regressions, effectively balancing the data and reducing confounding bias.

For potential outcome prediction, we use separate MLPs for treated and control groups to capture differing conditional outcome distributions. Given the network-integrated representations for latent confounding factors $\mathbf{r}_{c,i}$, adjustment factors $\mathbf{r}_{y,i}$, and peer

exposure g_i , the predictions are defined as:

$$\hat{y}_i(t_i, g_i) = \begin{cases} f_1(\mathbf{r}_{c,i}, \mathbf{r}_{y,i}, g_i), & t_i = 1, \\ f_0(\mathbf{r}_{c,i}, \mathbf{r}_{y,i}, g_i), & t_i = 0, \end{cases} \quad (11)$$

where f_1 and f_0 are outcome prediction functions implemented by separate MLPs for treated and control groups, respectively.

Additionally, we employ a treatment prediction network that concatenates $\mathbf{r}_{c,i}$ and instrumental factors $\mathbf{r}_{t,i}$ to predict t_i :

$$\hat{p}(t_i = 1 | \mathbf{r}_{c,i}, \mathbf{r}_{t,i}) = \sigma(g_t(\mathbf{r}_{c,i}, \mathbf{r}_{t,i})), \quad (12)$$

where $\sigma(\cdot)$ is the sigmoid function. This auxiliary task encourages the learned representations to capture treatment assignment patterns.

3.3 Causal Regularization Module

To ensure that the learned latent representations are disentangled in accordance with the underlying causal structure, we design a causal regularization module composed of two components.

3.3.1 Independence Constraint on Adjustment Factors. Since the pre-aggregation adjustment representation $\mathbf{z}_y$ is causally independent of the treatment assignment t , its distribution should be invariant across treated and control units. We quantify this using the distributional discrepancy:

$$D_{\text{adj}} = \text{IPM}(\{\mathbf{z}_{y,i}\}_{t_i=0}, \{\mathbf{z}_{y,j}\}_{t_j=1}), \quad (13)$$

where the integral probability metric (IPM) quantifies the discrepancy between the empirical distributions of $\mathbf{z}_y$ in the treated and control groups [35]. We implement the IPM through the Wasserstein-1 distance, $\text{Wass}(\cdot, \cdot)$, using an efficient approximation [7]. A smaller D_{adj} indicates stronger independence between the adjustment representation and treatment assignment, facilitating precise disentanglement of the adjustment factor.

3.3.2 Information Preservation Regularization. While the independence constraint reduces treatment-related leakage into $\mathbf{z}_y$, it is also essential to ensure that $\mathbf{z}_y$ indeed encodes meaningful adjustment information. To this end, we introduce an auxiliary prediction task that infers the observed outcome y_i from the network-integrated adjustment representation and treatment t_i :

$$\hat{y}_i = g_y([\mathbf{r}_{y,i}, t_i]), \quad (14)$$

where $g_y(\cdot)$ is a MLP and $[\cdot, \cdot]$ denotes concatenation. Because $\mathbf{r}_{y,i}$ aggregates neighborhood information from $\mathbf{z}_{y,i}$, improving the predictive accuracy of g_y implicitly encourages $\mathbf{z}_{y,i}$ to capture adjustment information.

Similarly, to promote concentration of instrumental information within the instrumental factor, we introduce a treatment prediction task:

$$\hat{p}(t_i = 1) = \sigma(g'_t(\mathbf{r}_{t,i})), \quad (15)$$

where $g'_t(\cdot)$ is a MLP and $\sigma(\cdot)$ is the sigmoid function. This objective encourages $\mathbf{z}_t$ to capture instrumental information while minimizing its spread into other factor representations.

3.4 Loss Function

We optimize the variational autoencoder by maximizing the Evidence Lower Bound (ELBO) conditioned on observed variables $\mathbf{u}$, with latent factors $\mathbf{z} = (\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y)$, $\mathcal{L}_{\text{ELBO}} = \frac{1}{n} \sum_i [\mathbb{E}_{q(\mathbf{z}_i | \mathbf{x}_i, \mathbf{u}_i)} [\log p(\mathbf{x}_i | \mathbf{z}_i)]$

$-\text{KL}(q(\mathbf{z}_i | \mathbf{x}_i, \mathbf{u}_i) \parallel p(\mathbf{z}_i | \mathbf{u}_i))]$. To mitigate confounding bias, we adopt a weighted mean squared error loss for outcome regression, $\mathcal{L}_y = \frac{1}{n} \sum_i w_i (y_i - \hat{y}_i)^2$, where w_i is obtained via density-ratio estimation (Eq. 10) using a binary classifier with loss $\mathcal{L}_w$. Treatment assignment is modeled with binary cross-entropy loss $\mathcal{L}_t = -\frac{1}{n} \sum_i [t_i \log \hat{p}(t_i) + (1 - t_i) \log(1 - \hat{p}(t_i))]$. We further impose causal regularization $\mathcal{L}_{\text{causal}} = \lambda_{\text{Wass}} \mathcal{L}_{\text{Wass}} + \lambda_{\text{info}} \mathcal{L}_{\text{info}}$ to enforce independence of adjustment factors and preserve predictive information.

The final objective is $\mathcal{L} = -\mathcal{L}_{\text{ELBO}} + \lambda_y \mathcal{L}_y + \lambda_t \mathcal{L}_t + \mathcal{L}_{\text{causal}}$, optimized alternately with $\mathcal{L}_w$ to ensure accurate ITE estimation. A detailed analysis of the computational complexity of all components is provided in our repository².

3.5 Theoretical Analysis

In this section, we provide theoretical insights into our model, including the identifiability of latent representations and individual treatment effects, as well as a generalization error bound on standard individual treatment effect estimation under network interference. Proofs of all theorems are available at the repository².

3.5.1 Identifiability of VAE. We aim to identify three types of latent factors: instrumental, confounding, and adjustment, and denote $\mathbf{z} = [\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y]$ with total dimension $d_z = D_t + D_c + D_y$. The generative model is defined as $p_{\theta}(\mathbf{x}, \mathbf{z} | \mathbf{u}) = p_{\mathbf{H}, \lambda}(\mathbf{z} | \mathbf{u}) \cdot p_{\mathbf{f}}(\mathbf{x} | \mathbf{z})$, where $p_{\mathbf{f}}(\mathbf{x} | \mathbf{z}) = p_{\epsilon}(\mathbf{x} - \mathbf{f}(\mathbf{z}))$ with injective function $\mathbf{f} : \mathbb{R}^{d_z} \rightarrow \mathbb{R}^{D_x}$ and $\epsilon \sim p_{\epsilon}(\epsilon)$ an independent noise variable. The latent prior is specified as $p_{\mathbf{H}, \lambda}(\mathbf{z} | \mathbf{u}) = \prod_{i=1}^{d_z} \frac{Q_i(z_i)}{Z_i(\mathbf{u})} \exp\left(\sum_{j=1}^k H_{i,j}(z_i) \lambda_{i,j}(\mathbf{u})\right)$, where Q_i is the base measure, Z_i is the normalizing constant, $\mathbf{H} = (H_{i,j})$ denotes the sufficient statistics, and $\lambda = (\lambda_{i,j})$ is the parameter function conditioned on $\mathbf{u}$.

Definition 3.1 (Identifiability). The model is *identifiable* if, for any two parameter sets $\theta = (\mathbf{f}, \mathbf{H}, \lambda)$ and $\theta' = (\mathbf{f}', \mathbf{H}', \lambda')$, $p_{\theta}(\mathbf{x} | \mathbf{u}) = p_{\theta'}(\mathbf{x} | \mathbf{u})$ implies the existence of an invertible matrix $\mathbf{M} \in \mathbb{R}^{d_z \times d_z}$ and a vector $\mathbf{c} \in \mathbb{R}^{d_z}$ such that, for all $\mathbf{x}$, $\mathbf{H}(\mathbf{f}^{-1}(\mathbf{x})) = \mathbf{M}\mathbf{H}'(\mathbf{f}'^{-1}(\mathbf{x})) + \mathbf{c}$.

This means that, in the deterministic case (i.e., $p_{\mathbf{f}}(\mathbf{x} | \mathbf{z}) = \delta(\mathbf{x} - \mathbf{f}(\mathbf{z}))$), $\mathbf{f}^{-1}(\mathbf{x})$ recovers $\mathbf{z}$ up to invertible linear transformations and pointwise nonlinearities, and in the noisy case, the posterior of $\mathbf{z}$ remains identifiable under mild conditions.

THEOREM 3.2. *Suppose the observed data are generated according to the generative process described above, and the following conditions hold:*

- (1) *The function $\mathbf{f}$ is injective.*
- (2) *The set $\{\mathbf{x} \in \mathcal{X} | \varphi_{\epsilon}(\mathbf{x}) = 0\}$, where φ_{ϵ} is the characteristic function of the noise distribution p_{ϵ} , has Lebesgue measure zero.*
- (3) *For each $i \in \{1, \dots, d_z\}$, the sufficient statistics $\{H_{i,j}\}_{j=1}^k$ are differentiable almost everywhere and linearly independent on any subset of $\mathcal{X}$ of positive measure.*

²<https://github.com/fandi-xkeai/NDiVAE>

(4) There exist $d_z k + 1$ distinct auxiliary inputs $\mathbf{u}^{(0)}, \dots, \mathbf{u}^{(d_z k)}$ such that the matrix

$$\mathbf{L} = [\lambda(\mathbf{u}^{(1)}) - \lambda(\mathbf{u}^{(0)}), \dots, \lambda(\mathbf{u}^{(d_z k)}) - \lambda(\mathbf{u}^{(0)})]$$

is invertible.

Then, the model parameters $\theta = (\mathbf{f}, \mathbf{H}, \lambda)$ are identifiable.

These conditions are commonly adopted in identifiable VAE frameworks to obtain theoretical guarantees [19]. Specifically, the injectivity of the generative mapping $\mathbf{f}$ ensures that distinct latent factors correspond to distinct observations, thereby making the parameters $(\mathbf{f}, \mathbf{H}, \lambda)$ identifiable. In practice, although the true generative process may not necessarily be injective, relaxing this assumption would lead to substantially more complex analysis, which is beyond the scope of this work. With identifiability of $\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y$, the identifiability of individual treatment effects in networked settings can be established.

Assumption 4 (Networked Unconfoundedness) [10]. The treatment assignment t_i , peer exposure g_i , and potential outcomes $y_i(t, g)$ are conditionally independent given the latent factors $\mathbf{z}_{c,i}$ and those of its neighbors $\{\mathbf{z}_{c,j}\}_{j \in \mathcal{N}_i}$. That is, all confounding factors are assumed to be captured by the local and neighborhood latent confounding factors, i.e., $y_i(t, g) \perp\!\!\!\perp t_i, g_i | \mathbf{z}_{c,i}, \{\mathbf{z}_{c,j}\}_{j \in \mathcal{N}_i}$.

THEOREM 3.3. Under Assumptions 1–4 and Theorem 3.2, the individual treatment effect τ_i in Eq. (2) is identifiable.

3.5.2 Generalization Error Bound for ITE Estimation. We follow the analysis framework of [4, 34] to establish a generalization error bound for ITE estimation under network interference.

Let $\mathcal{Z}_c, \mathcal{Z}_y$ and $\mathcal{Z}_c^N, \mathcal{Z}_y^N$ denote individual and neighborhood latent confounding and adjustment factors' spaces. We define the representation mapping function $\phi_{cy} : \mathcal{Z}_c \times \mathcal{Z}_c^N \times \mathcal{Z}_y \times \mathcal{Z}_y^N \rightarrow \mathcal{R}_c \times \mathcal{R}_y$, such that the concatenated representation $\mathbf{r}_{cy} = \phi_{cy}(\mathbf{z}_c, \mathbf{z}_c^N, \mathbf{z}_y, \mathbf{z}_y^N) = (\phi_c(\mathbf{z}_c, \mathbf{z}_c^N), \phi_y(\mathbf{z}_y, \mathbf{z}_y^N)) = (\mathbf{r}_c, \mathbf{r}_y)$ where ϕ_c and ϕ_y map latent confounding and adjustment factors (including neighborhood information) into their respective representation spaces $\mathcal{R}_c$ and $\mathcal{R}_y$. The inverse mapping is denoted by ψ_{cy} . For notational simplicity, we omit the unit index i and define $\tilde{\mathbf{z}}_{cy} := \mathcal{Z}_c \times \mathcal{Z}_c^N \times \mathcal{Z}_y \times \mathcal{Z}_y^N$. Then, we denote $\tilde{\mathbf{z}}_{cy} \in \tilde{\mathcal{Z}}_{cy}$.

Let $L : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}^+$ be a pointwise loss function between the true and predicted outcomes. We define the **per-unit loss** for the tuple $(\tilde{\mathbf{z}}_{cy}, t, g)$ as:

$$\ell_{f, \phi_{cy}}(\tilde{\mathbf{z}}_{cy}, t, g) = \int_{\mathcal{Y}} L(y(t, g), f(\phi_{cy}(\tilde{\mathbf{z}}_{cy}), t, g)) p(y(t, g) | \tilde{\mathbf{z}}_{cy}) dy(t, g),$$

where f is the outcome prediction function. Unlike some prior works [4, 16, 34] that define losses on observed covariates $\mathbf{x}$, we formulate them on latent factors $\tilde{\mathbf{z}}_{cy}$; see the repository² for details.

Definition 3.4 (Average Factual and Counterfactual Loss). Given the per-unit loss $\ell_{f, \phi_{cy}}(\tilde{\mathbf{z}}_{cy}, t, g)$, the *average factual loss* is defined as:

$$\epsilon_F = \mathbb{E}_{p(t, g)} \mathbb{E}_{p(\tilde{\mathbf{z}}_{cy} | t, g)} \ell_{f, \phi_{cy}}(\tilde{\mathbf{z}}_{cy}, t, g), \quad (16)$$

and the *average counterfactual loss* is defined as:

$$\epsilon_{CF} = \mathbb{E}_{p(t, g)} \mathbb{E}_{p(t', g')} \mathbb{E}_{p(\tilde{\mathbf{z}}_{cy} | t', g')} \ell_{f, \phi_{cy}}(\tilde{\mathbf{z}}_{cy}, t, g). \quad (17)$$

Definition 3.5 (Weighted Average Factual Loss). Let $w(\tilde{\mathbf{z}}_{cy}, t, g)$ be a *weighting function* corresponding to the density ratio between

the calibration and observed distributions. The *weighted average factual loss* is defined as:

$$\begin{aligned} \epsilon_F^{(w)} &= \mathbb{E}_{p(t, g)} \int_{\tilde{\mathcal{Z}}_{cy}} \ell_{f, \phi_{cy}}(\tilde{\mathbf{z}}_{cy}, t, g) w(\tilde{\mathbf{z}}_{cy}, t, g) p(\tilde{\mathbf{z}}_{cy} | t, g) d\tilde{\mathbf{z}}_{cy} \\ &= \mathbb{E}_{p(t, g, \tilde{\mathbf{z}}_{cy})} [w(\tilde{\mathbf{z}}_{cy}, t, g) \ell_{f, \phi_{cy}}(\tilde{\mathbf{z}}_{cy}, t, g)]. \end{aligned} \quad (18)$$

Definition 3.6. Let $\tilde{\mathbf{z}}_{cy} \in \mathcal{Z}_c \times \mathcal{Z}_c^N \times \mathcal{Z}_y \times \mathcal{Z}_y^N$ denote the concatenated latent factors of a target unit, and let $\mathbf{r}_{cy} = \phi_{cy}(\tilde{\mathbf{z}}_{cy})$ be its representation. For treatment-neighborhood pairs (t, g) and (t', g') , the true individual treatment effect is defined as Eq. (2):

$$\tau_{(t, g), (t', g')}(\tilde{\mathbf{z}}_{cy}) := m(t, g; \tilde{\mathbf{z}}_{cy}) - m(t', g'; \tilde{\mathbf{z}}_{cy}),$$

where $m(t, g; \tilde{\mathbf{z}}_{cy}) = \mathbb{E}[y(t, g) | \tilde{\mathbf{z}}_{cy}]$. Given a hypothesis f operating on the representation space, the estimated ITE is

$$\hat{\tau}_{(t, g), (t', g')}(\tilde{\mathbf{z}}_{cy}) := f(\phi_{cy}(\tilde{\mathbf{z}}_{cy}), t, g) - f(\phi_{cy}(\tilde{\mathbf{z}}_{cy}), t', g').$$

The error in treatment effect estimation is evaluated by the expected Precision in Estimation of Heterogeneous Effect (PEHE), defined as:

$$\begin{aligned} \epsilon_{PEHE}(f, \phi_{cy}) &= \mathbb{E}_{p(t, g) p(t', g')} \int_{\tilde{\mathcal{Z}}_{cy}} (\hat{\tau}_{(t, g), (t', g')}(\tilde{\mathbf{z}}_{cy}) \\ &\quad - \tau_{(t, g), (t', g')}(\tilde{\mathbf{z}}_{cy}))^2 p(\tilde{\mathbf{z}}_{cy}) d\tilde{\mathbf{z}}_{cy}. \end{aligned} \quad (19)$$

THEOREM 3.7. Let $\phi_{cy} : \tilde{\mathcal{Z}}_{cy} \rightarrow \mathcal{R}_c \times \mathcal{R}_y$ be a one-to-one representation mapping with inverse ψ_{cy} , where $\tilde{\mathcal{Z}}_{cy} := \mathcal{Z}_c \times \mathcal{Z}_c^N \times \mathcal{Z}_y \times \mathcal{Z}_y^N$. Let $\tilde{\mathbf{z}}_{cy}$ denote the concatenated latent factors for the target unit and $\mathbf{r}_{cy} = \phi_{cy}(\tilde{\mathbf{z}}_{cy})$. Let $f : \mathcal{R}_c \times \mathcal{R}_y \times \mathcal{T} \times \mathcal{G} \rightarrow \mathcal{Y}$ be an outcome hypothesis. Let $\mathcal{S}$ be a family of functions $s : \tilde{\mathcal{Z}}_{cy} \times \mathcal{T} \times \mathcal{G} \rightarrow \mathbb{R}$. Assume there exists a constant $B_{\phi_{cy}} > 0$ such that for all $(t, g) \in \mathcal{T} \times \mathcal{G}$, $\frac{1}{B_{\phi_{cy}}} \ell_{f, \phi_{cy}}(\tilde{\mathbf{z}}_{cy}, t, g) \in \mathcal{S}$. Define the transported function class on representation space as $\mathcal{S}_{\phi} := \{\bar{s}(\mathbf{r}_{cy}, t, g) = s(\psi_{cy}(\mathbf{r}_{cy}), t, g) \mid s \in \mathcal{S}\}$. Then the average counterfactual loss admits the bound

$$\epsilon_{CF} \leq \epsilon_F^{(w)} + B_{\phi_{cy}} \text{IPM}_{\mathcal{S}_{\phi}}(p_{\phi_{cy}}(\mathbf{r}_{cy}) p(t, g), w(\psi_{cy}(\mathbf{r}_{cy}), t, g) p(t, g, \mathbf{r}_{cy})).$$

where $p_{\phi_{cy}}(\mathbf{r}_{cy})$ and $p(t, g, \mathbf{r}_{cy})$ denote the marginal and joint push-forward densities induced by ϕ_{cy} , respectively. The IPM for a function class $\mathcal{V}$ is defined as $\text{IPM}_{\mathcal{V}}(p, q) := \sup_{v \in \mathcal{V}} |\int v(x) (p(x) - q(x)) dx|$.

THEOREM 3.8. Assume the conditions of the preceding theorem hold, and let the pointwise loss L be the squared loss. Denote by σ_Y the variance of $y(t, g)$ under the distribution $p(t, g) p(\tilde{\mathbf{z}}_{cy})$, i.e. $\sigma_Y := \mathbb{E}_{p(t, g) p(\tilde{\mathbf{z}}_{cy})} [\text{Var}(y(t, g) | \tilde{\mathbf{z}}_{cy})]$. Then the PEHE satisfies

$$\begin{aligned} \epsilon_{PEHE}(f, \phi_{cy}) &\leq 4\epsilon_F^{(w)} + 4B_{\phi_{cy}} \text{IPM}_{\mathcal{S}_{\phi}}(p_{\phi_{cy}}(\mathbf{r}_{cy}) p(t, g), \\ &\quad w(\psi_{cy}(\mathbf{r}_{cy}), t, g) p(t, g, \mathbf{r}_{cy})) - 4\sigma_Y. \end{aligned} \quad (20)$$

Under our model design, the reweighting function is specified as Eq. (10), i.e., $w_i = \frac{p_{\text{cal}}(t_i, g_i, \mathbf{r}_{c,i})}{p_{\text{obs}}(t_i, g_i, \mathbf{r}_{c,i})} = \frac{p(t_i, g_i) p(\mathbf{r}_{c,i})}{p(t_i, g_i, \mathbf{r}_{c,i})}$, which corresponds to choosing $w(\tilde{\mathbf{z}}_{cy}, t, g) = \frac{p_{\phi_{cy}}(\mathbf{r}_{cy}) p(t, g)}{p(t, g, \mathbf{r}_{cy})} \Big|_{\mathbf{r}_{cy} = \phi_{cy}(\tilde{\mathbf{z}}_{cy})}$. This alignment of distributions eliminates the integral probability metric term, $\text{IPM}_{\mathcal{S}_{\phi}} = 0$, thus establishing $\epsilon_{CF} = \epsilon_F^{(w)}$ exactly. Consequently, the confounding bias is removed in the reweighted representation, and the ϵ_{PEHE} upper bound in Theorem 3.8 reduces to a function of the weighted factual loss, which can be directly optimized to minimize the ϵ_{PEHE} . Referring back to our loss design in Section 3.4,

by controlling the variance to keep the negative ELBO non-negative, minimizing the total loss $\mathcal{L}$ effectively reduces the ϵ_{PEHE} upper bound, enabling reliable individual treatment effect estimation.

4 Experiments

In this section, we evaluate NDiVAE's performance. We first describe the experimental settings, and then present an analysis and comparison of the results with baseline models.

4.1 Experimental Settings

4.1.1 Datasets. This section details the datasets and data generation process used to evaluate our model. Due to the infeasibility of obtaining ground-truth individual treatment effects, we employ both fully synthetic and widely used semi-synthetic datasets based on real-world social networks.

Synthetic Data. We generate fully synthetic datasets following the procedure illustrated in Figure 1. An undirected, unweighted, self-loop-free graph $G = (V, E)$ with $n = 1000$ nodes is sampled from an Erdős-Rényi model $G(n, p)$, where $p = \frac{d}{n-1}$, with expected degree $d = 6$. The adjacency matrix $A \in \{0, 1\}^{n \times n}$ is symmetric with zero diagonal. For each node i , latent factors $z_{t,i}, z_{c,i}, z_{y,i} \sim \mathcal{N}(0, 1)$, and an auxiliary variable $u_i = w_t z_{t,i} + w_c z_{c,i} + w_y z_{y,i} + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma_u^2)$, are sampled with fixed coefficients w_t, w_c and w_y , with $\sigma_u = 0.1$.

Observed node features $\mathbf{x}_i \in \mathbb{R}^{D_x}$ with feature dimension $D_x = 10$ are generated by a linear transformation of latent factors with additive Gaussian noise:

$$\mathbf{x}_i = W_x^\top [z_{t,i}, z_{c,i}, z_{y,i}]^\top + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2 \mathbf{I}), \quad (21)$$

where $W_x \in \mathbb{R}^{3 \times D_x}$ is a randomly generated matrix with entries $(W_x)_{i,j} \sim \mathcal{N}(1.0, 0.5^2)$ and $\sigma = 0.1$.

Treatment propensity is computed from own and neighbors' latent factors: $s_{t,i} = \sigma(z_{t,i})$, $s_{c,i} = \sigma(z_{c,i})$, $\bar{s}_{t,i} = \frac{1}{|N(i)|} \sum_{j \in N(i)} s_{t,j}$, $\bar{s}_{c,i} = \frac{1}{|N(i)|} \sum_{j \in N(i)} s_{c,j}$, and $\ell_{t,i} = \gamma_t s_{t,i} + \gamma_c s_{c,i} + \gamma_{nt} \bar{s}_{t,i} + \gamma_{nc} \bar{s}_{c,i}$, where $\sigma(\cdot)$ is the sigmoid function and all parameters $\gamma_t, \gamma_c, \gamma_{nt}, \gamma_{nc}$ are set to 1.0. The treatment probability is $p_{t,i} = \sigma(\ell_{t,i})$, and treatment is assigned by thresholding at the mean propensity $\bar{p}_t$, i.e., $t_i = 1\{p_{t,i} > \bar{p}_t\}$.

Outcomes are generated accounting for latent confounding factors, adjustment factors, treatment effects, peer exposure, and noise:

$$\begin{aligned} u_{c,i} &= \sigma(z_{c,i}), u_{y,i} = \sigma(z_{y,i}), \bar{u}_{c,i} = \frac{1}{|N(i)|} \sum_{j \in N(i)} u_{c,j}, \bar{u}_{y,i} = \frac{1}{|N(i)|} \sum_{j \in N(i)} u_{y,j}, g_i = \frac{1}{|N(i)|} \sum_{j \in N(i)} t_j, \text{ and} \\ \ell_{y,i} &= \delta_c u_{c,i} + \delta_y u_{y,i} + \delta_{nc} \bar{u}_{c,i} + \delta_{ny} \bar{u}_{y,i}, \\ y_i &= \beta_t t_i + \beta_g g_i + \ell_{y,i} + t_i \cdot (\delta_y u_{y,i} + \delta_c u_{c,i} + \delta_{ny} \bar{u}_{y,i} + \delta_{nc} \bar{u}_{c,i}) + \beta_\epsilon \epsilon_i, \end{aligned} \quad (22)$$

where all parameters $\delta_c, \delta_y, \delta_{nc}, \delta_{ny}, \beta_t, \beta_g, \beta_\epsilon$ are set to 1.0 and $\epsilon_i \sim \mathcal{N}(0, 1)$.

Semi-Synthetic Datasets. *BlogCatalog (BC)* is a blogging network with nodes as bloggers and edges as social links; *Flickr* is a social platform with nodes as users and edges as social relationships. Following prior works [16], node features are projected through Latent Dirichlet Allocation (LDA) [3] into 10 dimensions and decomposed into three latent factors, namely instrumental $\mathbf{z}_t \in \mathbb{R}^2$,

confounding $\mathbf{z}_c \in \mathbb{R}^2$, and adjustment $\mathbf{z}_y \in \mathbb{R}^2$, as well as auxiliary variables $\mathbf{u} \in \mathbb{R}^4$. Similarly, observed features, treatments, and outcomes are generated following the models described above for both homogeneous ($y_i = \beta_t t_i + \beta_g g_i + \ell_{y,i} + \beta_\epsilon \epsilon_i$) and heterogeneous settings (Eq. (22)) [5], with different parameter values, resulting in four datasets: BC (homo), Flickr (homo), BC (hete), and Flickr (hete). Following [16], each network is partitioned into training, validation, and test sub-networks via METIS [18], with results reported as "within sample" (training) and "without sample" (test).

4.1.2 Metrics. We adopt two metrics for evaluation. The first is the Mean Absolute Error (MAE) on average effects (AME, ASE, ATCE), defined as $\epsilon_{\text{average}} = \left| \frac{1}{n} \sum_{i=1}^n \hat{\tau}_i - \frac{1}{n} \sum_{i=1}^n \tau_i \right|$, where τ_i and $\hat{\tau}_i$ denote the ground-truth and estimated treatment effects for unit i , respectively. The second is the Root Precision in Estimation of Heterogeneous Effect (RPEHE) for individual effects (IME, ISE, ITCE), given by $\epsilon_{\text{RPEHE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{\tau}_i - \tau_i)^2}$. Lower values indicate better performance for both metrics.

4.1.3 Baselines. We compare NDiVAE with baselines for independent observational data (TARNet [34], CFR [34], TEDVAE [28]) extended with peer-exposure g as TARNet+ g , CFR+ g and TEDVAE+ g , and network-aware methods (NetDeconf [11], TNDGVA [9]) augmented with g as NetDeconf+ g and TNDGVA+ g . NetEst [16] explicitly models interference using GNNs.

4.1.4 Implementation Details. NDiVAE is implemented in PyTorch and run on an NVIDIA RTX 3080Ti GPU³. All parameters are randomly initialized and optimized via Adam [20] with full-batch training. For the BC and Flickr datasets, we train for 40 epochs with a learning rate of 10^{-3} , a decay of 0.01, and a weight decay of 10^{-4} . We conduct ten independent runs with different random seeds and report the mean and one standard deviation for all evaluation metrics. Both the encoder and decoder consist of 4 layers with 256 units per layer, and the latent dimensions for $\mathbf{z}_t, \mathbf{z}_c$, and $\mathbf{z}_y$ are set to 2. Loss weights $\lambda_y, \lambda_t, \lambda_{\text{Wass}}$, and λ_{info} are selected via grid search $\{0.01, 0.1, 1, 10, 100\}$. For synthetic datasets, training is 200 epochs with latent dimensions 1, hidden dimension 64, 3 layers. Each synthetic dataset is partitioned into 60%/20%/20% for training, validation, and testing, and all experiments are repeated five times with different random seeds.

4.2 Results Analysis

4.2.1 Empirical Validation of Latent Factor Identifiability. To empirically validate the identifiability guarantees established by Theorem 3.2, we conduct experiments on a fully synthetic dataset. Figure 3 presents correlation heatmaps between the learned latent factors $\mathbf{z}_t, \mathbf{z}_c, \mathbf{z}_y$ and the true latent factors $\mathbf{z}_t, \mathbf{z}_c$, and $\mathbf{z}_y$ on both train and test sets. The distinct diagonal patterns in these heatmaps indicate a strong correspondence between the learned and true latent factors, demonstrating that our method effectively recovers the latent causal factors. This finding provides empirical support that the estimated latent factors are identifiable up to permutation and component-wise invertible transformations, thereby corroborating the theoretical claims of Theorem 3.2.

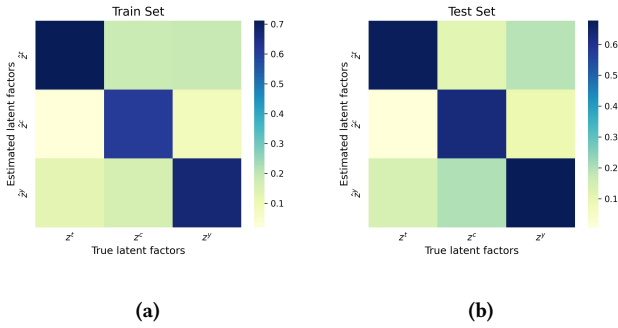
³The code is publicly available on this repository².

Table 1: Experimental results of causal effect estimation on the BC (homo) dataset. The best result is highlighted in bold.

metric	setting	effect	TNRNet+g	CFR+g	TEDVAE+g	NetDeconf+g	TNDGVA+g	NetEst	NDiVAE_W	NDiVAE_R	NDiVAE
$\epsilon_{\text{average}}$	within sample	AME	0.1662 $\pm$ 0.1070	0.0938 $\pm$ 0.0642	0.0881 $\pm$ 0.0586	0.1259 $\pm$ 0.1129	0.0937 $\pm$ 0.1187	0.2153 $\pm$ 0.3547	0.0512 $\pm$ 0.0357	0.0485 $\pm$ 0.0266	0.0338 $\pm$ 0.0233
		ASE	0.3177 $\pm$ 0.1347	0.2467 $\pm$ 0.1275	0.2159 $\pm$ 0.1288	0.3182 $\pm$ 0.1541	0.0989 $\pm$ 0.1085	0.1497 $\pm$ 0.1132	0.0266 $\pm$ 0.0369	0.0228 $\pm$ 0.0178	0.0174 $\pm$ 0.0111
		ATCE	0.2614 $\pm$ 0.2033	0.2880 $\pm$ 0.1919	0.1392 $\pm$ 0.1342	0.4126 $\pm$ 0.2781	0.1554 $\pm$ 0.2178	0.3450 $\pm$ 0.4728	0.0662 $\pm$ 0.0311	0.0570 $\pm$ 0.0337	0.0358 $\pm$ 0.0271
	without sample	AME	0.2771 $\pm$ 0.1855	0.1639 $\pm$ 0.1467	0.1157 $\pm$ 0.0818	0.5170 $\pm$ 0.7240	0.1917 $\pm$ 0.3411	0.2123 $\pm$ 0.3536	0.0477 $\pm$ 0.0393	0.0598 $\pm$ 0.0502	0.0311 $\pm$ 0.0210
		ASE	0.3182 $\pm$ 0.1344	0.2476 $\pm$ 0.1334	0.2206 $\pm$ 0.1290	0.3303 $\pm$ 0.1553	0.1116 $\pm$ 0.1136	0.1470 $\pm$ 0.1205	0.0275 $\pm$ 0.0364	0.0242 $\pm$ 0.0188	0.0226 $\pm$ 0.0123
		ATCE	0.2811 $\pm$ 0.1470	0.2904 $\pm$ 0.1812	0.1354 $\pm$ 0.1310	0.6731 $\pm$ 0.5610	0.2405 $\pm$ 0.3452	0.3697 $\pm$ 0.4577	0.0583 $\pm$ 0.0350	0.0694 $\pm$ 0.0554	0.0351 $\pm$ 0.0215
ϵ_{RPEHE}	within sample	IME	2.1624 $\pm$ 2.9761	1.3525 $\pm$ 2.0256	0.1318 $\pm$ 0.0495	0.2311 $\pm$ 0.2142	0.4949 $\pm$ 0.7659	0.2940 $\pm$ 0.3418	0.0891 $\pm$ 0.0484	0.0947 $\pm$ 0.0566	0.0755 $\pm$ 0.0229
		ISE	0.3479 $\pm$ 0.1317	0.2775 $\pm$ 0.1264	0.2491 $\pm$ 0.1272	0.3485 $\pm$ 0.1566	0.1244 $\pm$ 0.1190	0.2128 $\pm$ 0.1684	0.0542 $\pm$ 0.0391	0.0500 $\pm$ 0.0224	0.0397 $\pm$ 0.0174
		ITCE	2.2030 $\pm$ 2.9522	1.4401 $\pm$ 1.9848	0.1986 $\pm$ 0.1161	0.4823 $\pm$ 0.3054	0.5202 $\pm$ 0.7855	0.4075 $\pm$ 0.4647	0.1013 $\pm$ 0.0382	0.1059 $\pm$ 0.0516	0.0809 $\pm$ 0.0289
	without sample	IME	3.3259 $\pm$ 5.3978	1.9884 $\pm$ 4.1848	0.1517 $\pm$ 0.0812	2.4238 $\pm$ 4.9257	0.8381 $\pm$ 1.7804	1.5389 $\pm$ 2.7632	0.0890 $\pm$ 0.0455	0.1028 $\pm$ 0.0734	0.0729 $\pm$ 0.0226
		ISE	0.3492 $\pm$ 0.1296	0.2790 $\pm$ 0.1304	0.2561 $\pm$ 0.1270	0.3621 $\pm$ 0.1548	0.1533 $\pm$ 0.1162	1.1862 $\pm$ 2.0066	0.0507 $\pm$ 0.0382	0.0502 $\pm$ 0.0238	0.0439 $\pm$ 0.0194
		ITCE	3.3869 $\pm$ 5.3543	2.0598 $\pm$ 4.1560	0.2025 $\pm$ 0.1110	2.5329 $\pm$ 4.8607	0.8618 $\pm$ 1.7624	1.6045 $\pm$ 2.5149	0.0958 $\pm$ 0.0402	0.1154 $\pm$ 0.0664	0.0758 $\pm$ 0.0227

Table 2: Experimental results of causal effect estimation on the Flickr (homo) dataset. The best result is highlighted in bold.

metric	setting	effect	TNRNet+g	CFR+g	TEDVAE+g	NetDeconf+g	TNDGVA+g	NetEst	NDiVAE_W	NDiVAE_R	NDiVAE
$\epsilon_{\text{average}}$	within sample	AME	0.3301 $\pm$ 0.2671	0.1736 $\pm$ 0.2014	0.0644 $\pm$ 0.0499	0.1392 $\pm$ 0.204	0.6282 $\pm$ 1.6108	0.2724 $\pm$ 0.3061	0.109 $\pm$ 0.0689	0.0942 $\pm$ 0.0825	0.0747 $\pm$ 0.0379
		ASE	0.4529 $\pm$ 0.2208	0.4577 $\pm$ 0.2347	0.2964 $\pm$ 0.1996	0.5735 $\pm$ 0.2516	0.2084 $\pm$ 0.2719	0.3904 $\pm$ 0.2679	0.0555 $\pm$ 0.0427	0.0527 $\pm$ 0.0358	0.0458 $\pm$ 0.0385
		ATCE	0.3199 $\pm$ 0.2927	0.4382 $\pm$ 0.3408	0.1097 $\pm$ 0.0874	0.67 $\pm$ 0.4161	0.6129 $\pm$ 1.3377	0.6481 $\pm$ 0.5575	0.0659 $\pm$ 0.0574	0.0933 $\pm$ 0.0842	0.0499 $\pm$ 0.0242
	without sample	AME	0.3691 $\pm$ 0.4896	0.1102 $\pm$ 0.0646	0.0677 $\pm$ 0.0511	0.9335 $\pm$ 1.4515	0.3437 $\pm$ 0.4291	0.2409 $\pm$ 0.2876	0.1057 $\pm$ 0.0696	0.0957 $\pm$ 0.0776	0.0761 $\pm$ 0.0444
		ASE	0.4559 $\pm$ 0.2185	0.4585 $\pm$ 0.2309	0.2874 $\pm$ 0.1969	0.5802 $\pm$ 0.2498	0.2291 $\pm$ 0.2595	0.3796 $\pm$ 0.2647	0.0554 $\pm$ 0.0426	0.0505 $\pm$ 0.034	0.0432 $\pm$ 0.0369
		ATCE	0.5373 $\pm$ 0.3695	0.5104 $\pm$ 0.3575	0.1095 $\pm$ 0.0882	1.0589 $\pm$ 1.0363	0.4893 $\pm$ 0.6894	0.6444 $\pm$ 0.5131	0.0645 $\pm$ 0.0568	0.0957 $\pm$ 0.0863	0.0418 $\pm$ 0.0191
ϵ_{RPEHE}	within sample	IME	5.96 $\pm$ 12.152	4.0843 $\pm$ 9.2523	0.1538 $\pm$ 0.054	0.6731 $\pm$ 1.6054	1.3972 $\pm$ 2.7996	0.4718 $\pm$ 0.5644	0.127 $\pm$ 0.0651	0.1163 $\pm$ 0.069	0.0952 $\pm$ 0.0361
		ISE	0.4814 $\pm$ 0.2167	0.4881 $\pm$ 0.2316	0.3317 $\pm$ 0.1997	0.6107 $\pm$ 0.2459	0.2278 $\pm$ 0.2758	0.5515 $\pm$ 0.5781	0.0754 $\pm$ 0.0387	0.079 $\pm$ 0.0303	0.0651 $\pm$ 0.0331
		ITCE	6.0323 $\pm$ 12.1126	4.3247 $\pm$ 9.1438	0.1919 $\pm$ 0.0944	1.1937 $\pm$ 1.4563	1.4148 $\pm$ 2.6422	0.8075 $\pm$ 0.7548	0.0945 $\pm$ 0.0513	0.1138 $\pm$ 0.0747	0.0762 $\pm$ 0.0303
	without sample	IME	3.06 $\pm$ 4.0431	1.1369 $\pm$ 1.4098	0.1617 $\pm$ 0.0576	2.7954 $\pm$ 3.4841	0.8059 $\pm$ 0.7056	0.7519 $\pm$ 0.8659	0.126 $\pm$ 0.0642	0.1156 $\pm$ 0.0673	0.0964 $\pm$ 0.0382
		ISE	0.492 $\pm$ 0.2141	0.4959 $\pm$ 0.2277	0.3277 $\pm$ 0.1969	0.626 $\pm$ 0.241	0.2708 $\pm$ 0.2583	0.902 $\pm$ 0.7967	0.0779 $\pm$ 0.0389	0.0772 $\pm$ 0.0298	0.0614 $\pm$ 0.0312
		ITCE	3.2132 $\pm$ 3.9217	1.4038 $\pm$ 1.2814	0.2024 $\pm$ 0.0968	3.0451 $\pm$ 3.1392	0.9054 $\pm$ 0.8493	1.0487 $\pm$ 0.8195	0.0956 $\pm$ 0.0485	0.1141 $\pm$ 0.0783	0.0677 $\pm$ 0.0234

**Figure 3: Heatmaps illustrating the correlation between learned latent factors z and true latent factors (including z_t , z_c , and z_y) on both training and testing datasets for the synthetic data.**

4.2.2 Results Comparison on Real-World Datasets. We compare NDiVAE against representative baselines from both independent observational settings and those incorporating network structure. Tables 1 and 2 summarize the experimental results on the BC (homo), Flickr (homo)⁴. Several key observations can be made:

⁴Results for the BC (hete) and Flickr (hete) datasets can be found in the same repository².

- Across all six evaluation metrics, including both average and individual treatment effect estimation, and all datasets, NDiVAE consistently demonstrates superior performance, often with a substantial margin over both independent-data and network-based baselines. Occasional instances where one or two baselines slightly outperform NDiVAE are limited to specific metrics and do not reduce its overall advantage.
- Exploiting network structure significantly improves performance. Methods that ignore graph information (e.g., CFR+g, TARNet+g, TEDVAE+g) are generally outperformed by approaches that leverage network information (e.g., NetDeconf+g, TNDGVA+g, NetEst), emphasizing the importance of capturing relational dependencies when working with networked observational data.
- Disentanglement-based methods (TEDVAE+g, TNDGVA+g) achieve competitive performance, highlighting the importance of disentangling instrumental, confounding, and adjustment factors. Nevertheless, NDiVAE outperforms them by jointly modeling multi-factor disentanglement and network interference within a unified variational framework.
- The consistent and robust performance of NDiVAE across both homogeneous and heterogeneous outcome settings demonstrates its adaptability to diverse data generating mechanisms, establishing it as a flexible and effective solution for causal effect estimation in networked data.

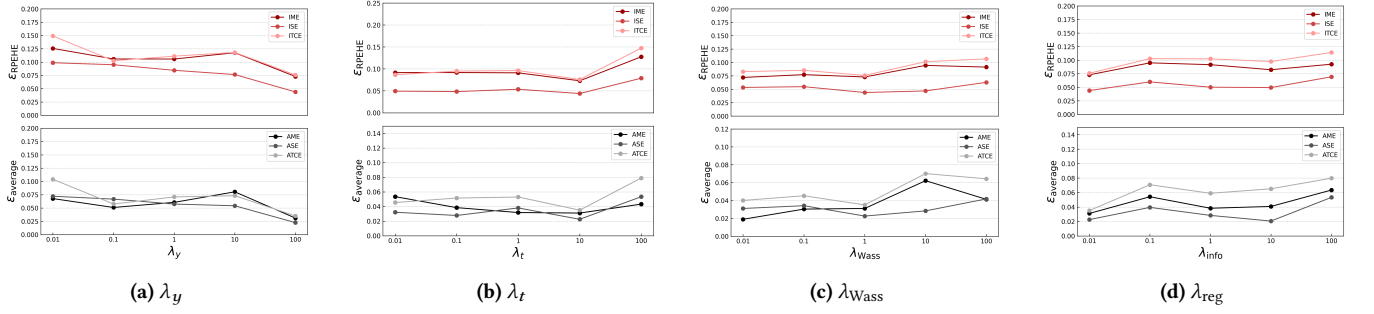


Figure 4: Sensitivity analysis of different hyperparameters on model performance. Each subplot shows the influence of one hyperparameter on the evaluation metric.

4.3 Ablation Study

We conduct ablation experiments on the BC (homo) and Flickr (homo) datasets, shown in Tables 1 and 2, comparing the full NDiVAE model with two variants⁵: NDiVAE_W (without sample weighting) and NDiVAE_R (without the causal regularization module). Across all metrics and datasets, the complete NDiVAE consistently outperforms these variants. This confirms that the sample weighting mechanism plays a significant role in mitigating confounding bias, enhancing model robustness. Similarly, the causal constraint regularization effectively guides the model to learn disentangled latent factors that better capture the underlying causal structure, leading to improved treatment effect estimation accuracy.

4.4 Hyperparameter Analysis

We conduct a sensitivity analysis on four key hyperparameters: λ_y , λ_t , λ_{Wass} , and λ_{info} , using the BC (homo) dataset. Figure 4 shows their impact on individual treatment effect metrics (top row) and average treatment effect metrics (bottom row). Results indicate that increasing λ_y consistently enhances performance, highlighting the importance of accurate outcome prediction. Performance is stable across λ_t and λ_{Wass} , with optimal values at 10 and 1, respectively. Excessively large λ_t degrades performance by placing excessive weight on treatment prediction, while too high λ_{Wass} weakens results by overemphasizing disentanglement of $\mathbf{z}_y$, reducing predictive accuracy. The independence regularization λ_{info} remains stable, with best results at 0.01. Overall, the model is robust across different hyperparameter settings.

5 Related Work

Causal effect estimation on independent observational data has been widely studied under SUTVA. Tree-based ensemble models

such as BART [6] and causal forests [36] provided flexible non-parametric estimators. Representation-learning methods including CFR [34] learned balanced feature spaces to reduce covariate shift, while disentanglement-based approaches such as DRCFR [12] and DeR-CFR [37] separated latent factors to better control confounding bias. Generative models including TEDVAE [39] and EDVAE [27] further used encoders to estimate potential outcome distributions.

⁵Results for BC (hete) and Flickr (hete) are provided in our repository².

Overall, these methods assume independent units and overlook dependencies induced by network structures.

For networked observational data, early studies also relied on SUTVA. NetDeconf [11] employed graph-based representation learning with IPM regularization to balance confounders and reduce bias. Subsequent work introduced disentanglement into this setting. GDC [13] and TNDGVA [9] disentangled latent factors on graphs to improve ITE estimation. However, these models are not suitable when spillover effects are present. To tackle network interference, Liu et al. [26] developed generalized inverse probability-weighted and Hájek-type estimators, while Forastiere et al. [10] extended propensity score adjustments to incorporate neighborhood treatments. Graph neural network-based approaches, including NetEst [16], HyperSCI [29], and SITE [24], exploited network structure and neighbor aggregation for scalable estimation. Further, Chen et al. [5] designed a doubly robust estimator under interference, and Zhao et al. [41] addressed heterogeneous interference via dual weighting regression. Despite these advances, most methods treat all observed features as sources of confounding, without explicitly disentangling instrumental, confounding, and adjustment factors, thereby limiting accurate potential outcome estimation and ITE prediction. Our work builds upon these frameworks but differs from prior methods by explicitly modeling distinct latent factors and leveraging disentanglement to more effectively estimate causal effects under network interference.

6 Conclusion

In this paper, we propose NDiVAE, a novel framework designed to estimate individual treatment effects from networked observational data under network interference. NDiVAE leverages an identifiable VAE to disentangle each unit's latent representation into three distinct factors, including instrumental, confounding, and adjustment factors. Independent graph aggregation networks are then used to integrate neighborhood information, resulting in network-integrated representations for each factor at each node. To mitigate confounding bias, a sample reweighting regression approach is employed for learning potential outcomes, while causal regularization ensures precise learning of the latent factors. We further provide theoretical guarantees, including identifiability of the VAE and a generalization error bound for treatment effect estimation, demonstrating the theoretical validity of the proposed framework. Extensive experiments on synthetic and semi-synthetic datasets demonstrate the effectiveness of NDiVAE and its superiority over existing methods.

For future work, it would be worthwhile to apply the NDiVAE framework to real-world networked observational datasets to provide stronger evidence of its practical applicability and robustness. Additionally, extending the framework to account for latent factors that cannot be captured by observational proxy variables is an important direction for future research.

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Ethical Considerations

This work presents methodological advances in estimating individual treatment effects in the presence of network interference. Although the proposed framework is technical, its application in real-world domains such as healthcare or the social sciences may involve fairness and privacy considerations, especially when observational data include biased or sensitive information. Practitioners are advised to evaluate fairness, ensure data privacy, and consider the broader implications when applying the method in practice.

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